Structural complexity of spoken languages – the significance of modularity

Structural complexity in spoken language refers to the complexity of the **linguistic structures used when people produce and understand speech**.

It concerns how words, phrases, clauses, sounds, and meanings are **organized and related to one another**.

Structural complexity = the degree to which linguistic units are hierarchically organized and interconnected across phonological, morphological, syntactic, semantic, and discourse levels.

Complexity in spoken language

Utterance	Speaker's Intention	speech act
"The window is open.“	Simply state a fact.	assertation
"The window is open!“	Warn someone that cold air is coming in.	warning
"The window is open.“	Indirectly ask someone to close it.	request
"The window is open.“	Blame someone for leaving it open.	Complaint
"The window is open.“	Explain why the room is cold.	Explanation
"The window is open?“	Seek confirmation.	Question

Structural module of language

- Phonological module; sound structure(phonology, syllable rules)
- Morphological module ; word formation (prefix, suffixes, inflection)
- Syntactic module; sentence structure and word order
- Semantic module ; meaning at the word and sentence level.
- Pragmatic module; Meaning in context, discourse interpretation

Example-Modular Interaction

- I have been teaching Linguistics since 2024.

Balancing complexity

- She knows that he left.(introduce a factive proposition)

(*That* signals assertion)

She wonders if he left- implies uncertainty , making an embedded question.

Example- Structural Dependence

- The boy who was standing near the gate saw the dog that was chasing the cat.
- John came because he wanted to meet the professor who had invited him to the conference.

Causal/ purpose

- He stayed home because he was sick.(because expresses causality)
- He left early for the meeting to start on time.(for encodes goal or intention, linking to modality.

Semantic / pragmatics

- She left.(neutral)
- She left?(surprise, doubt)

Implicature

- A. are you coming to the party?
- B.I have an exam tomorrow.

Syntax –pragmatics simplicity

- IMPLICATURE
- I think (that) she is right-----assertation
- He said (that) he was leaving
- Pragmatic ensures assertation.
- **Ambiguity**
- She said that she will leave**.(assertion)**
- She said if she will leave.(**question)**

Summing up

- Modular units accounts for both structural diversity and universal constraints in spoken language.
- Modularity balancing local syntactic rules with global interpretative flexibility.